AI COMMUNICATION SYSTEMS STUDY KIT

BEFORE / AFTER ANALYSIS

A comparison of manual learning and a simulation-supported study workflow for AM and FM.

Student: _____

Department: Electronics and Communication Engineering (ECE)

College: _____

Date: _____

Note: This is a qualitative project analysis template. It does not claim measured improvements or real user-test results. Add your own observations if you conduct a test.

1. Purpose

This analysis describes how a student's learning workflow may differ before and after using the AI Communication Systems Study Kit, which includes explanations, AM/FM simulation graphs, a glossary, quiz questions, and supporting documentation.

2. Before and After Comparison

Aspect	Before using the study kit	After using the study kit
Concept learning	May rely mainly on textbook explanations or lecture notes.	Can review short definitions and structured explanations alongside the topic.
AM/FM waveform understanding	Waveform behavior may be difficult to imagine from equations alone.	Can inspect generated plots and compare message, carrier, AM, and FM signals.
AM vs FM comparison	Differences may be scattered across notes.	Can use a side-by-side comparison of what changes and what remains constant.
Practice and revision	May need to create revision questions separately.	Can use the included glossary, quiz, and viva questions for practice.
Repeated graph creation	Graphs may need to be plotted and saved manually each time.	A Python script can generate and save graph files in one run, if executed successfully.
Progress evidence	Understanding may not be recorded in a consistent format.	Can use a checklist and record quiz scores or observations to track progress.

3. Signal-Specific Before / After Analysis

Amplitude Modulation (AM)

Before: A student may find it hard to connect the equation to the waveform envelope. After: Viewing the AM plot can help the student observe that the carrier amplitude follows the message signal. The modulation index and parameter values should be checked to avoid overmodulation in a standard AM example.

Frequency Modulation (FM)

Before: It may be difficult to visualize frequency changes from the equation alone. After: Viewing the FM plot can help the student observe that the spacing between cycles varies with the message while the ideal carrier amplitude remains constant.

4. How to Measure Actual Improvement

To make this a real before/after evaluation, ask the same learners to complete the same short quiz before and after using the kit. Record the actual scores; do not invent values.

Measure	Before	After
Quiz score (out of 10)	_____ / 10	_____ / 10
Time to explain AM (minutes)	_____	_____
Time to explain FM (minutes)	_____	_____
Confidence rating (1–5)	_____	_____
Common mistake observed	_____	_____

5. Analysis and Limitations

The study kit is designed to make revision more structured and to provide visual support through simulated signals. Whether it improves learning must be established using actual learner feedback or pre-test/post-test scores. Graph generation is a programmed simulation/automation feature; call it an AI feature only if a real AI component is implemented and tested.

6. Conclusion

The before/after comparison suggests that combining concise explanations, simulation graphs, and quiz practice can make AM/FM study more organized and visual. A measured conclusion about learning improvement should be written only after collecting real results.

Student notes / observations:

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